Extending class imbalance mitigation benchmarks to more datasets

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Abstract

A prior study benchmarked class-imbalance mitigation methods on TCGA-UT, a pan-cancer whole-slide-image cohort with a native, moderately imbalanced slide distribution (Queisler, 2026). This report extends the same benchmark machinery to two additional datasets, keeping the frozen Virchow2 representation, a bounded per-slide evidence budget, and the patient-disjoint evaluation protocol fixed, and asks whether the method comparison remains meaningful when the data regime changes. The datasets are chosen to bracket the imbalance spectrum: BRACS, an organ-specific H&E breast-lesion cohort with a mildly imbalanced native seven-subtype distribution (head-to-tail ≈2.3:1), and CAMELYON16, a sentinel lymph-node metastasis cohort whose mask-derived tumor/normal patch distribution is severely imbalanced (≈39:1). Both patch-feature and whole-slide-image feature-bag method families are compared, each within its own prediction regime, and a plain cross-entropy no-mitigation baseline (CE, MIL CE) is added so mitigation gains are measured against an unweighted reference. On BRACS, CFAL leads the patch regime while most other methods—including ProGAN augmentation, which falls slightly below CE—are statistically indistinguishable from the unweighted baseline, and in the WSI-bag regime only focal loss and SC-MIL clearly exceed a competitive MIL CE baseline; both regimes remain substantially miscalibrated after temperature scaling. On CAMELYON16, despite an imbalance an order of magnitude more severe, every method in both regimes reaches near-ceiling discrimination (balanced accuracy 0.96–0.99) and near-ideal calibration, and no mitigation method separates from the unweighted baseline. The two datasets together show that imbalance severity alone predicts neither task difficulty nor the value of mitigation: what governs the headroom for imbalance mitigation under a frozen-feature benchmark is the separability of the classes in the representation, not the head-to-tail ratio.

Contents

1 Introduction

A companion study compared class-imbalance mitigation methods under the native, moderately imbalanced slide distribution of TCGA-UT (head-to-tail ≈28:1) across 32 cancer types (Queisler, 2026; Tomczak et al., 2015; Komura et al., 2022). That benchmark fixes a frozen Virchow2 encoder, a bounded per-slide evidence budget, and patient-disjoint splits, so that the only thing being compared is how each mitigation method reshapes minibatch exposure or the per-example loss. The natural next question is external validity: does the same benchmark machinery stay meaningful when it is pointed at a different dataset, with a different organ, a different label set, and a different native class distribution?

This report takes that step by extending the benchmark to two further datasets on their *native* class distributions, chosen so that they bracket the imbalance spectrum from mild to severe. The design is deliberately conservative: the representation family (frozen Virchow2), the bounded per-slide evidence budget, the patient-disjoint split protocol, the method families, and the validation-based model-selection rule are all carried over unchanged, and only the dataset is swapped. A dataset therefore acts as a check on how far the benchmark generalizes rather than as a new experimental variable.

The first dataset is BRACS, an H&E breast-pathology dataset with 547 WSIs and 4,539 regions of interest (ROIs) from 189 patients, annotated by consensus of three board-certified pathologists into three lesion types and seven subtypes (Brancati et al., 2022). The atypical categories are clinically important and comparatively scarce, so BRACS is a genuine class-imbalance setting without any resampling, but its native head-to-tail ratio (≈2.3:1) is only mild. The second dataset is CAMELYON16, an organ-specific cohort of sentinel lymph-node sections for breast-cancer metastasis detection (Ehteshami Bejnordi et al., 2017; Litjens et al., 2018). Its pixel-level tumor annotations make it possible to label individual tiles as tumor or normal, and because metastatic tissue is scarce within a section, the resulting native patch distribution is severely imbalanced (≈39:1). Together the two datasets test whether the method comparison holds both where imbalance is mild and where it is severe, and they differ from TCGA-UT and from each other in organ, label granularity, and annotation type—exactly the variation that stresses a benchmark's generality.

Both regimes from the companion study are carried over. Patch-feature methods include cross-entropy, class weighting, balanced sampling, focal loss, Scholz-style metric losses, CFAL, OKO set learning, and ProGAN synthetic augmentation (Scholz et al., 2024). WSI-bag methods include cross-entropy, class weighting, balanced sampling, focal loss, RankMix, SC-MIL, and MDE-MIL (Chen and Lu, 2023; Juyal et al., 2024; Ling et al., 2025). Patch and WSI-bag methods are reported separately because they consume different prediction units and answer regime-specific questions.

2 Datasets

The benchmark evaluates each dataset on its *native* class distribution: the observed per-class support is measured after a patient-disjoint preparation and used directly, without any resampling. This report covers two datasets chosen to span the imbalance spectrum: BRACS, whose seven breast-lesion subtypes are only mildly imbalanced, and CAMELYON16, whose patch-level tumor/normal distribution is severely imbalanced. Further datasets can be added under the same protocol.

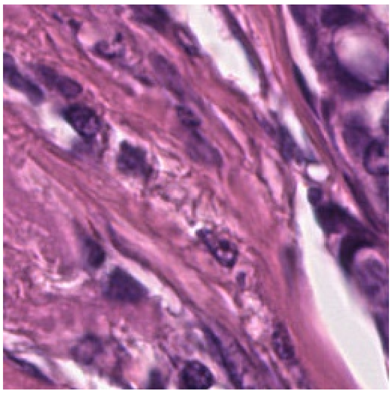
2.1 BRACS

BRACS is an organ-specific H&E breast-lesion dataset rather than a pan-cancer dataset, and its annotations are lesion-subtype labels rather than cancer-type labels. It contains 547 WSIs and 4,539 ROIs from 189 patients, annotated by consensus of three board-certified pathologists (Brancati et al., 2022). The labels form three lesion types—benign, atypical, and malignant—and seven subtypes: normal tissue (N), pathological benign (PB), usual ductal hyperplasia (UDH), flat epithelial atypia (FEA), atypical ductal hyperplasia (ADH), ductal carcinoma in situ (DCIS), and invasive carcinoma (IC).

The benchmark uses the seven-subtype target because it is the level at which class imbalance is most relevant. The atypical categories FEA and ADH are clinically important precancerous lesions and are less common than many benign or malignant samples, so collapsing BRACS to the three lesion types would hide part of the rare-class problem. Patient identifiers from the BRACS summary sheet define the split unit, and the prepared benchmark enforces patient-disjoint train, validation, and test partitions across three seeds—matching the leakage-control standard used for TCGA-UT.

BRACS evidence is derived from the ROI set. Each annotated ROI is tiled deterministically into 256×256 RGB patches, capped at 30 tiles per ROI to match the evidence budget used for the companion study. Patch methods consume one frozen Virchow2 feature per ROI tile (Vorontsov et al., 2024; Zimmermann et al., 2024), represented as the concatenation of the CLS token and the mean of the patch tokens (2,560-dimensional vectors). The WSI-bag pipeline consumes grouped ROI-tile features as bounded bags, preserving the same patch-versus-bag comparison while avoiding an additional tissue-detection step on full WSIs. This choice prioritizes label fidelity: all training instances originate from pathologist-selected lesion regions rather than from unlabeled background tissue.

After patch extraction, 387 of the 547 published WSIs contribute at least one annotated ROI tile; the remaining slides have no tiled ROI annotations available. The prepared benchmark covers 151 patients, the full 4,539 ROI metadata records, and 98,282 extracted patches (Table 1). The seven-subtype WSI support spans PB (196 slides) at the head to DCIS (87 slides) at the tail, a native head-to-tail ratio of approximately 2.25:1 (Figure 1). This is mild imbalance relative to the pan-cancer TCGA-UT cohort, but it reflects the true clinical distribution of breast-lesion subtypes and is used as-is.



Property	Value
Patients	151
WSIs with ROI tiles	387
ROI metadata rows	4,539
ROI tiles	98,282
Subtype labels	N, PB, UDH, FEA, ADH, DCIS, IC
WSI support range	87–196
WSI imbalance ratio	$\approx$2.3:1

Table 1: BRACS dataset properties used by the native benchmark, after patient-disjoint preparation and ROI tiling, alongside an example 256×256 ROI tile. Each tile is one frozen Virchow2 feature consumed by the patch regime and a candidate instance in the WSI bag.

2.2 CAMELYON16

CAMELYON16 is an organ-specific H&E dataset of sentinel lymph-node sections for the detection of breast-cancer metastases (Ehteshami Bejnordi et al., 2017; Litjens et al., 2018). It differs from BRACS along the axes that matter for a generality check. Its label is binary—each slide is either *normal* or contains *tumor* (metastasis)—and its expert annotations are supplied as pixel-level masks that mark tumor and normal tissue, rather than as region-of-interest crops. Holding

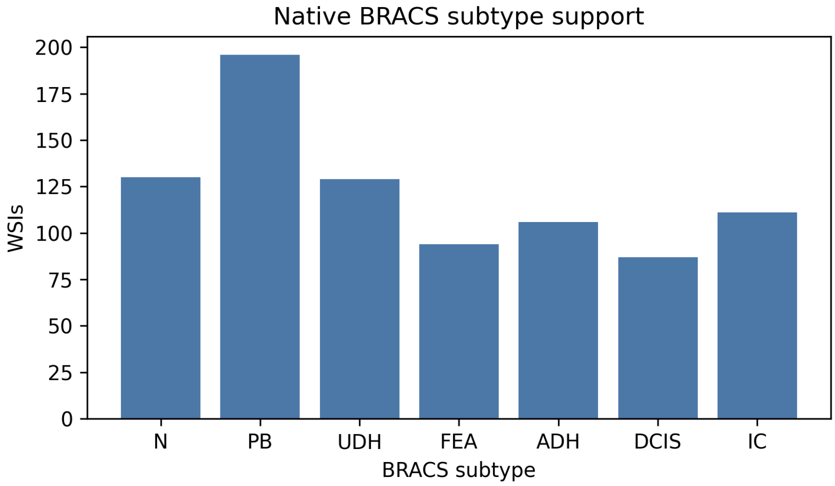


Figure 1: Native BRACS WSI support by seven-subtype label after patient-disjoint preparation. The distribution is used directly as the benchmark's native class-imbalance setting.

the frozen Virchow2 representation and the patient-disjoint protocol fixed while changing the organ, the label granularity, and, above all, the severity of the native imbalance, CAMELYON16 is the deliberate severe counterpart to BRACS's mild seven-subtype distribution.

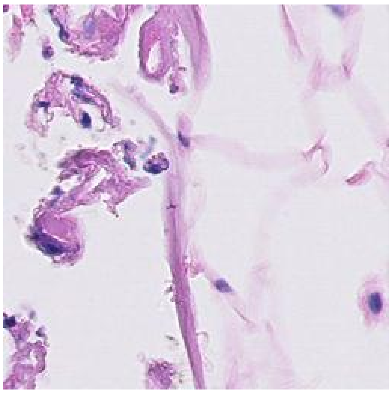
The benchmark reuses the 398 slides for which pre-extracted 256×256 tissue tiles at $20\times$ magnification are available, out of the 399 published slides, comprising 160 tumor and 238 normal slides. Each slide is its own patient (case), so slide-disjoint train, validation, and test partitions across three seeds coincide with patient-disjoint splits. CAMELYON16 is evaluated in the same two regimes as BRACS, but with regime-specific labels. In the patch regime, each tile is labelled tumor or normal by intersecting it with the published tumor mask; because metastatic regions occupy only a small fraction of the sectioned tissue, tumor tiles are scarce and the native patch distribution is severely imbalanced (head-to-tail $\approx$39:1; Table 2 and Figure 2). In the WSI-bag regime, each slide is a single bag carrying its slide-level tumor/normal label, which recovers the canonical CAMELYON16 metastasis-detection task.

Two dataset-specific choices follow from this design. First, the 20 tumor slides whose annotations are not exhaustive—so that unannotated tumor could be mislabelled as normal—are excluded from the patch regime but retained in the WSI-bag regime, where only the reliable slide-level label is used. Second, the per-slide evidence budget is set data-driven rather than fixed: the WSI-bag size is the empirical median number of available tissue tiles per slide, which for CAMELYON16 is 7,874 tiles. This is two orders of magnitude larger than the 30-tile budget that the tile-sparse TCGA-UT and BRACS cohorts admit, and reflects that a whole lymph-node section yields far more tissue than a single lesion ROI. Patch evidence is otherwise the same frozen Virchow2 2,560-dimensional embedding used throughout the benchmark (Vorontsov et al., 2024; Zimmermann et al., 2024), computed once per tile and shared by both regimes.

3 Methods

The method taxonomy follows the companion benchmark (Queisler, 2026). Cross-entropy is the no-mitigation reference. Data-level methods change minibatch exposure through balanced sampling or synthetic augmentation. Algorithm-level methods change the per-example loss through class weighting, focal loss, metric-derived objectives, prototype-affinity objectives, or OKO set supervision. WSI-specific methods change how slide bags are represented or mixed.

Full method definitions follow Queisler (2026) and Scholz et al. (2024). RankMix, SC-MIL,



Property	Value
WSIs (tumor / normal)	160 / 238
Patch-labelled WSIs	378
WSI-bag size (median tiles)	7,874
Patch labels	tumor, normal
Tumor / normal patches	59,672 / 2,340,040
Patch imbalance ratio	≈39.2:1

Table 2: CAMELYON16 dataset properties used by the native benchmark, after patient-disjoint preparation and mask-based patch labelling, alongside an example 256×256 tumor tile. The WSI-bag size is the data-driven median of available $20\times$ tissue tiles per slide.

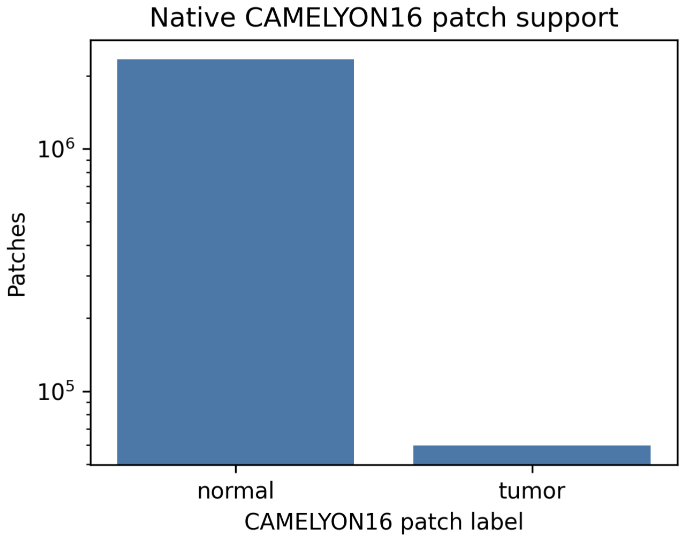


Figure 2: Native CAMELYON16 patch support by tumor/normal label after patient-disjoint preparation and mask-based labelling (log scale). Tumor tiles are roughly 39 times scarcer than normal tiles, giving the benchmark its severe native imbalance setting.

and MDE-MIL are WSI-bag methods whose mechanisms operate on bags rather than isolated patch embeddings (Chen and Lu, 2023; Juyal et al., 2024; Ling et al., 2025). The plain cross-entropy baselines (CE for the patch regime, MIL CE for the WSI-bag regime) apply uniform class weights and no resampling, so any improvement from a mitigation method is measured relative to doing nothing.

4 Evaluation Protocol

The benchmark evaluates each dataset on its native class distribution. Hyperparameters are selected independently for every method: for each candidate configuration the validation macro F1 is averaged over the three patient-disjoint seeds, and the configuration with the highest mean is chosen, with ties broken by mean validation balanced accuracy. A single configuration is therefore fixed per method (shared across seeds). Selected configurations are evaluated on the fixed test split, and results are reported as mean ± standard deviation across seeds. Macro F1 is the primary discrimination metric because it weights classes equally; balanced accuracy

Family	Methods	Regime
No-mitigation baseline	CE, MIL CE	Patch, WSI bag
Data-level sampling and augmentation	Balanced sampling, RankMix	Patch, WSI bag
Algorithm-level losses	Weighted CE, Focal loss, CFAL	Patch (CFAL); Patch and WSI bag (others)
Hybrid sampling–loss	CE + soft F1, CE + soft MCC, OKO, MDE-MIL	Patch (first three, OKO); WSI bag (MDE-MIL)
Synthetic data generation	ProGAN augmentation	Patch
Representation and architecture	SC-MIL	WSI bag

Table 3: Method families evaluated on both native benchmarks (BRACS and CAMELYON16), following the taxonomy of Queisler (2026). CE and MIL CE are unweighted cross-entropy baselines, added so mitigation gains are measured against a genuine no-mitigation reference. Patch and WSI-bag methods are compared only within their own input regime.

and accuracy are reported as complementary discrimination metrics.

The comparison is intentionally regime-specific. Patch-feature methods are compared only with other patch-feature methods, and WSI-bag methods are compared only with other WSI-bag methods. The two regimes share the patient-disjoint splits, the frozen Virchow2 representation family, and a bounded per-slide evidence budget, but they differ in prediction unit and aggregation mechanism. The evidence budget itself is dataset-specific: BRACS caps evidence at 30 tiles per ROI, matching TCGA-UT, whereas CAMELYON16 sets the WSI-bag size to the empirical median of 7,874 available tiles per slide, so the budget tracks how much tissue each cohort actually provides.

Calibration is evaluated alongside discrimination using expected calibration error (ECE; Naeini et al., 2015; Guo et al., 2017). For each method and seed, a single temperature is fit by minimizing negative log likelihood on the validation split and applied unchanged to the test score matrix before softmax (Guo et al., 2017). Tables 6 and 7 report ECE before and after temperature scaling with the fitted temperature.

5 Results

Each dataset is evaluated on its own native distribution, with the patch-feature and WSI-bag regimes reported separately. Scores are specific to each dataset's task and are not directly comparable across datasets or to the 32-class TCGA-UT benchmark, because the label set, task difficulty, and cohort size all differ. The two datasets turn out to occupy opposite corners of the difficulty–imbalance plane—BRACS is mildly imbalanced but hard, CAMELYON16 severely imbalanced but easy—and it is this contrast, rather than either dataset alone, that tests the benchmark's generality.

5.1 BRACS

The patch-feature and WSI-bag regimes are reported separately (Tables 4 and 5). The absolute discrimination and calibration numbers are markedly lower than on TCGA-UT for two concrete reasons: fine-grained breast-lesion subtyping (distinguishing e.g. FEA, ADH, and DCIS) is intrinsically harder than pan-cancer type classification, and the native breast-only training pool is far smaller than the large constructed pan-cancer pools used for TCGA-UT, so each method has both a harder decision boundary and less data to fit it. This is consistent with the external literature: even recent task-specific BRACS models—purpose-built graph and transformer architectures trained end-to-end for this task—report only mid-60% weighted F1 on seven-subtype ROI classification (HACT-Net 61.5%, ScoreNet 64.4%), with the atypical UDH and ADH subtypes still around 45% F1 (Pati et al., 2022; Stegmüller et al., 2023). A frozen-feature benchmark under a bounded evidence budget should therefore not be expected to approach TCGA-UT-level discrimination; the binding limitation is fine-grained lesion morphology and limited task-specific representation learning, not class imbalance alone.

Method	Accuracy	Balanced accuracy	Macro F1
CFAL	0.581 ± 0.082	0.512 ± 0.050	0.499 ± 0.059
CE + soft F1	0.543 ± 0.059	0.490 ± 0.031	0.474 ± 0.035
OKO	0.527 ± 0.064	0.482 ± 0.034	0.466 ± 0.036
Weighted CE	0.523 ± 0.054	0.468 ± 0.036	0.458 ± 0.038
CE + soft MCC	0.519 ± 0.048	0.467 ± 0.022	0.452 ± 0.024
Balanced sampling	0.513 ± 0.053	0.458 ± 0.035	0.447 ± 0.035
Focal loss	0.504 ± 0.057	0.461 ± 0.032	0.446 ± 0.040
CE	0.515 ± 0.070	0.457 ± 0.025	0.443 ± 0.035
ProGAN augmentation	0.498 ± 0.045	0.455 ± 0.021	0.437 ± 0.023

Table 4: BRACS native patch-feature test results (accuracy, balanced accuracy, and macro F1) for validation-selected configurations, mean ± standard deviation across patient-disjoint seeds.

Method	Accuracy	Balanced accuracy	Macro F1
SC-MIL	0.633 ± 0.087	0.453 ± 0.063	0.472 ± 0.056
Focal loss	0.613 ± 0.070	0.469 ± 0.104	0.459 ± 0.060
MIL CE	0.603 ± 0.030	0.446 ± 0.043	0.430 ± 0.014
Weighted CE	0.577 ± 0.089	0.415 ± 0.049	0.423 ± 0.055
MDE-MIL	0.612 ± 0.087	0.426 ± 0.076	0.422 ± 0.064
RankMix	0.579 ± 0.127	0.408 ± 0.084	0.414 ± 0.083
Balanced sampling	0.617 ± 0.044	0.400 ± 0.061	0.395 ± 0.066

Table 5: BRACS native WSI-bag test results for validation-selected configurations, mean ± standard deviation across patient-disjoint seeds.

In the patch regime, CFAL leads on all three metrics (BAcc 0.512, F1 0.499), 2.2 points of balanced accuracy ahead of the next-best method (CE + soft F1, BAcc 0.490) and 5.5 points ahead of the unweighted CE baseline (BAcc 0.457, F1 0.443). The remaining methods—Weighted CE, focal loss, CE + soft MCC, and balanced sampling—cluster tightly between BAcc 0.458 and 0.468, indistinguishable from the CE baseline within one standard deviation (all $\sigma \approx 0.03$–0.08). OKO (BAcc 0.482) sits between this cluster and CFAL. ProGAN augmentation is the weakest method in the regime (BAcc 0.455, F1 0.437), fractionally below the unweighted CE baseline on every metric—unlike on the companion TCGA-UT benchmark, where ProGAN led at the mildest severity, BRACS's native imbalance never becomes severe enough for synthetic augmentation to pay off. The overall picture is that most patch mitigation methods provide little measurable lift over unweighted cross-entropy on BRACS's mild native imbalance, and one (ProGAN) falls slightly below it; only CFAL and, to a lesser extent, CE + soft F1 and OKO separate from the no-mitigation baseline.

In the WSI-bag regime, the unweighted MIL CE baseline (BAcc 0.446, F1 0.430) is competitive with most dedicated mitigation methods: it outperforms Weighted CE (BAcc 0.415), balanced sampling (BAcc 0.400), RankMix (BAcc 0.408), and MDE-MIL (BAcc 0.426) on balanced accuracy. Only focal loss (BAcc 0.469, the regime leader) and SC-MIL (BAcc 0.453, F1 0.472, the regime's highest macro F1) clearly exceed the CE baseline. Between-seed standard deviations are large throughout this regime (up to $\sigma \approx 0.10$–0.13 for RankMix and SC-MIL), so individual rankings should be read as numerical leads rather than statistically separated outcomes; the qualitative pattern—that several bag-composition and resampling methods do not clearly beat plain MIL CE on BRACS—is nonetheless the direct motivation for including the baseline. Balanced sampling is the weakest method in this regime (BAcc 0.400, F1 0.395), below the unweighted baseline on every metric.

Method	ECE	ECE+TS	T
CE	0.376 ± 0.058	0.056 ± 0.036	6.361 ± 1.353
Weighted CE	0.372 ± 0.044	0.047 ± 0.025	6.009 ± 1.212
Focal loss	0.280 ± 0.041	0.038 ± 0.029	3.300 ± 0.672
CFAL	0.418 ± 0.085	0.068 ± 0.032	0.058 ± 0.010
CE + soft F1	0.416 ± 0.054	0.078 ± 0.010	9.079 ± 0.906
CE + soft MCC	0.397 ± 0.045	0.058 ± 0.028	7.003 ± 1.266
Balanced sampling	0.378 ± 0.046	0.046 ± 0.028	6.069 ± 1.164
OKO	0.146 ± 0.055	0.043 ± 0.035	1.468 ± 0.312
ProGAN augmentation	0.394 ± 0.042	0.053 ± 0.031	6.337 ± 1.146

Table 6: BRACS native patch-feature calibration: ECE before (raw) and after (TS) temperature scaling, with the fitted temperature T, mean ± standard deviation across seeds.

Method	ECE	ECE+TS	T
MIL CE	0.329 ± 0.063	0.155 ± 0.002	3.301 ± 0.274
Weighted CE	0.332 ± 0.086	0.155 ± 0.034	3.313 ± 0.428
Focal loss	0.283 ± 0.073	0.134 ± 0.027	2.838 ± 0.395
Balanced sampling	0.292 ± 0.063	0.142 ± 0.030	3.156 ± 0.248
RankMix	0.152 ± 0.077	0.133 ± 0.054	1.160 ± 0.060
SC-MIL	0.259 ± 0.101	0.112 ± 0.033	3.126 ± 0.312
MDE-MIL	0.275 ± 0.043	0.152 ± 0.047	1.986 ± 0.242

Table 7: BRACS native WSI-bag calibration: ECE before and after temperature scaling, with the fitted temperature T. Columns as in Table 6.

Probability Calibration. In the patch regime, raw (pre-scaling) miscalibration is severe for most methods: ECE ranges from 0.28 to 0.42, requiring large temperature corrections ($T \approx 3$–9) to reach post-scaling ECE of 0.04–0.08. OKO is the exception, with by far the lowest raw ECE (0.146) and the smallest fitted temperature ($T \approx 1.47$), consistent with its behaviour on the companion TCGA-UT benchmark. CFAL again shows the CFAL-specific pattern of a near-degenerate low temperature ($T \approx 0.058$) alongside the highest post-scaling ECE (0.068) of any patch method, echoing its affinity-based outputs being nearly flat prior to scaling. Focal loss achieves the lowest post-scaling ECE (0.038). ProGAN augmentation calibrates similarly to the unweighted baselines (raw ECE 0.394, $T \approx 6.34$, post-scaling ECE 0.053), showing no distinctive miscalibration pattern despite its weaker discrimination. In the WSI-bag regime, calibration is uniformly worse than the patch regime and than the companion TCGA-UT benchmark:

post-scaling ECE ranges from 0.112 (SC-MIL, the best-calibrated method) to 0.155 (MIL CE and Weighted CE, tied for worst). RankMix has the lowest raw ECE (0.152) and smallest temperature ($T \approx 1.16$) of any WSI-bag method, but its post-scaling ECE (0.133) is not the best. The elevated residual miscalibration across all BRACS methods, relative to TCGA-UT, reflects the smaller training pool and the intrinsic difficulty of seven-class fine-grained subtyping rather than a failure specific to any one method.

5.2 CAMELYON16

On CAMELYON16 the picture inverts: the same frozen features that struggle with BRACS's fine-grained subtypes separate tumor from normal tissue almost perfectly, so despite an imbalance an order of magnitude more severe, every method reaches near-ceiling discrimination in both regimes (Tables 8 and 9).

Method	Accuracy	Balanced accuracy	Macro F1
CFAL	0.997 ± 0.002	0.961 ± 0.015	0.966 ± 0.008
Balanced sampling	0.997 ± 0.001	0.969 ± 0.014	0.960 ± 0.020
ProGAN augmentation	0.996 ± 0.001	0.956 ± 0.009	0.959 ± 0.019
CE	0.996 ± 0.001	0.962 ± 0.015	0.958 ± 0.020
Focal loss	0.996 ± 0.001	0.972 ± 0.010	0.957 ± 0.027
OKO	0.996 ± 0.000	0.974 ± 0.007	0.955 ± 0.034
CE + soft F1	0.996 ± 0.001	0.986 ± 0.002	0.952 ± 0.035
Weighted CE	0.996 ± 0.002	0.966 ± 0.012	0.952 ± 0.028
CE + soft MCC	0.995 ± 0.001	0.982 ± 0.006	0.940 ± 0.050

Table 8: CAMELYON16 native patch-feature test results (tumor vs. normal tiles) for validation-selected configurations, mean ± standard deviation across patient-disjoint seeds.

Method	Accuracy	Balanced accuracy	Macro F1
MDE-MIL	0.983 ± 0.017	0.984 ± 0.018	0.982 ± 0.018
SC-MIL	0.983 ± 0.017	0.984 ± 0.018	0.982 ± 0.018
Balanced sampling	0.977 ± 0.010	0.979 ± 0.012	0.977 ± 0.010
Focal loss	0.977 ± 0.010	0.979 ± 0.012	0.977 ± 0.010
RankMix	0.972 ± 0.020	0.974 ± 0.020	0.971 ± 0.020
MIL CE	0.966 ± 0.017	0.965 ± 0.021	0.965 ± 0.018
Weighted CE	0.966 ± 0.017	0.965 ± 0.021	0.965 ± 0.018

Table 9: CAMELYON16 native WSI-bag test results (slide-level metastasis detection) for validation-selected configurations, mean ± standard deviation across patient-disjoint seeds.

In the patch regime, balanced accuracy ranges from 0.961 (CFAL) to 0.986 (CE + soft F1) and macro F1 from 0.940 to 0.966, on 99.5–99.7% accuracy. The unweighted CE baseline (BAcc 0.962, F1 0.958) sits squarely inside this band, and no mitigation method separates from it by more than the between-seed noise: CFAL edges ahead on macro F1 (0.966) and CE + soft F1 on balanced accuracy (0.986), but the spread across the whole field is under 0.03 on either metric. The severe native imbalance (≈39:1) does not translate into a hard problem, because tumor and normal tiles are highly separable in the frozen Virchow2 space; there is little error left for any reweighting or resampling scheme to remove.

The WSI-bag regime tells the same story at the slide level. All seven methods score between balanced accuracy 0.965 and 0.984 and macro F1 0.965 and 0.982. MDE-MIL and SC-MIL lead (both F1 0.982), the unweighted MIL CE baseline (BAcc 0.965, F1 0.965, tied with Weighted CE) is again fully competitive, and the entire field is separated by less than 0.02 macro F1. Canonical slide-level metastasis detection is nearly solved from a bounded bag of Virchow2 features, leaving no meaningful gap for bag-composition or resampling methods to close.

Method	ECE	ECE+TS	T
CE	0.003 ± 0.001	0.001 ± 0.001	3.723 ± 0.142
Weighted CE	0.004 ± 0.002	0.001 ± 0.001	4.224 ± 0.503
Focal loss	0.003 ± 0.001	0.001 ± 0.000	4.547 ± 0.173
CFAL	0.464 ± 0.001	0.068 ± 0.004	0.050 ± 0.000
CE + soft F1	0.004 ± 0.001	0.001 ± 0.001	4.598 ± 0.179
CE + soft MCC	0.005 ± 0.001	0.002 ± 0.000	5.025 ± 0.191
Balanced sampling	0.003 ± 0.001	0.001 ± 0.001	4.071 ± 0.239
OKO	0.003 ± 0.000	0.001 ± 0.000	3.014 ± 0.100
ProGAN augmentation	0.003 ± 0.001	0.001 ± 0.001	4.119 ± 0.281

Table 10: CAMELYON16 native patch-feature calibration: ECE before (raw) and after (TS) temperature scaling, with the fitted temperature T, mean ± standard deviation across seeds.

Method	ECE	ECE+TS	T
MIL CE	0.027 ± 0.017	0.034 ± 0.017	0.050 ± 0.000
Weighted CE	0.027 ± 0.017	0.034 ± 0.017	0.050 ± 0.000
Focal loss	0.021 ± 0.013	0.023 ± 0.010	0.050 ± 0.000
Balanced sampling	0.022 ± 0.009	0.023 ± 0.010	0.050 ± 0.000
RankMix	0.022 ± 0.030	0.028 ± 0.024	0.273 ± 0.386
SC-MIL	0.020 ± 0.018	0.017 ± 0.017	0.050 ± 0.000
MDE-MIL	0.013 ± 0.008	0.017 ± 0.017	0.050 ± 0.000

Table 11: CAMELYON16 native WSI-bag calibration: ECE before and after temperature scaling, with the fitted temperature T. Columns as in Table 10.

Probability Calibration. Calibration follows discrimination: because the models are both confident and correct, they are well calibrated out of the box. In the patch regime raw ECE is 0.003–0.005 for every method except CFAL, and temperature scaling brings it to 0.001–0.002 (Table 10); the fitted temperatures ($T \approx$ 3–5) merely soften mildly over-confident logits. CFAL is again the lone exception, with the same near-degenerate low-temperature signature seen on BRACS (raw ECE 0.464, $T \approx 0.05$, post-scaling ECE 0.068). In the WSI-bag regime raw ECE is 0.013–0.027 and post-scaling ECE 0.017–0.034 (Table 11); MDE-MIL is the best calibrated (raw ECE 0.013), and for several methods the fitted temperature reaches the grid's lower floor ($T \approx 0.05$) without lowering ECE, reflecting how little miscalibration remains to correct on the small 398-slide cohort. Both regimes are markedly better calibrated than BRACS—patch post-scaling ECE around 0.001 versus BRACS's 0.04–0.08, and WSI-bag around 0.02–0.03 versus BRACS's 0.11–0.15—which is the calibration counterpart of the same separability gap.

6 Discussion

The benchmark's central design change from the companion TCGA-UT study is the addition of an unweighted no-mitigation baseline (CE, MIL CE), and the two regimes answer the question it was added to ask quite differently. In the patch regime, most mitigation methods do not clearly separate from the CE baseline on BRACS's mild native imbalance (≈2.3:1)—Weighted CE, focal loss, CE + soft MCC, and balanced sampling all sit within noise of unweighted cross-entropy, and ProGAN augmentation falls fractionally below it. Only CFAL, and to a lesser extent CE + soft F1 and OKO, produce a measurable gain. This is a plausible consequence of the imbalance itself being mild: with a native ratio an order of magnitude below TCGA-UT's ≈28:1, there may simply be less headroom for reweighting or resampling to help. ProGAN's reversal is particularly informative: on TCGA-UT it led at the mildest constructed severity before collapsing as tail-class support fell to single-digit patch counts at the harshest severity. BRACS's native imbalance never reaches that regime, so ProGAN never gets the tail-scarcity conditions under which synthetic augmentation would be expected to help, and it is left carrying only the risk of the generative step (distributional mismatch between synthetic and real patches) without the corresponding benefit.

The WSI-bag regime tells a sharper version of the same story: several mitigation methods (Weighted CE, balanced sampling, RankMix, MDE-MIL) score *below* the unweighted MIL CE baseline on balanced accuracy. Only focal loss and SC-MIL clearly improve on it. This is exactly the failure mode a no-mitigation baseline is meant to surface, and it would have been invisible in a benchmark that only compared mitigation methods against each other—the companion TCGA-UT report, which lacked this baseline, could not have detected it. Whether this reflects BRACS's smaller training pool (283 native training WSIs from 387 prepared slides, versus TCGA-UT's constructed pools of over 1000 slides), the milder native imbalance giving resampling less to correct, or a genuine difference in how bag-composition methods behave outside a pan-cancer setting cannot be distinguished from a single additional dataset; it is a hypothesis for the next dataset in this benchmark to test.

Calibration adds a second, independent axis on which BRACS is harder than TCGA-UT: even the best-calibrated method in each regime (focal loss for patch, SC-MIL for WSI-bag) needs temperature scaling to reach an ECE that is still 2–3× worse than typical TCGA-UT post-scaling values. This is consistent with the smaller training pool and finer-grained label set rather than a method-specific weakness, since the pattern holds across every method in both regimes.

CAMELYON16 sharpens the central lesson by inverting BRACS's regime. It is severely imbalanced (≈39:1, an order of magnitude beyond BRACS and comparable to TCGA-UT's ≈28:1) yet nearly solved: every method in both regimes exceeds 0.96 balanced accuracy, the unweighted baselines are competitive with the best mitigation method, and calibration is near-ideal without any dataset-specific tuning. Read together, the two datasets show that imbalance severity alone determines neither task difficulty nor the headroom available to mitigation. BRACS is mildly imbalanced but hard, because fine-grained lesion subtypes are poorly separated by the frozen features; CAMELYON16 is severely imbalanced but easy, because tumor and normal tissue are nearly separable in those same features. What governs whether a mitigation method can help is therefore the separability of the classes in the representation, not the head-to-tail ratio: a severely imbalanced but separable problem leaves a plain cross-entropy baseline essentially optimal, whereas a mildly imbalanced but entangled problem is exactly where the method choice, and the no-mitigation reference, come to matter. This is a direct caution against selecting or ranking imbalance-mitigation methods by imbalance ratio alone, and it is only visible because the benchmark now spans both corners of the difficulty–imbalance plane.

7 Conclusion

This report extended the class-imbalance mitigation benchmark introduced for TCGA-UT (Queisler, 2026) to two additional datasets on their native distributions, keeping the frozen Virchow2 representation, a bounded per-slide evidence budget, and the patient-disjoint evaluation protocol unchanged, and adding an unweighted cross-entropy baseline (CE, MIL CE) so mitigation gains are measured against doing nothing. The two datasets were chosen to bracket the imbalance spectrum. On BRACS's native seven-subtype distribution (≈2.3:1 head-to-tail), the patch regime shows CFAL leading by a clear margin (BAcc 0.512 vs. 0.457 for CE), with most other mitigation methods statistically indistinguishable from the no-mitigation baseline and ProGAN augmentation falling fractionally below it; in the WSI-bag regime several mitigation methods do not clearly improve on the unweighted MIL CE baseline, and only focal loss and SC-MIL do. On CAMELYON16's severely imbalanced patch distribution (≈39:1), by contrast, every patch and WSI-bag method reaches near-ceiling discrimination (balanced accuracy 0.96–0.99) and near-ideal calibration, and the unweighted baselines are competitive with the best mitigation method in both regimes.

Taken together, the two datasets show that method rankings are not universal and, more pointedly, that imbalance severity alone does not predict whether mitigation helps: CAMELYON16 is far more imbalanced than BRACS yet leaves essentially no headroom for mitigation, because its classes are separable in the frozen representation, whereas BRACS's milder imbalance coincides with a genuinely hard, entangled task where the method choice and the no-mitigation reference matter. What governs the value of imbalance mitigation under a frozen-feature benchmark is therefore class separability, not the head-to-tail ratio—a finding that only becomes visible with a native additional dataset, a genuine no-mitigation baseline, and coverage of both corners of the difficulty–imbalance plane. Further datasets can be added under the same protocol to test how far this pattern generalizes.

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